Worksheet 9.1

Identifying and classifying minerals

Student: ..... Class: .....

Identifying minerals

1. Two common rocks are composed of the minerals shown in the images at right. Limestone contains calcite, and sandstone contains quartz.



Calcite



Quartz

The following table provides information about the properties of these minerals.

Mineral	Colour	Streak (colour of powdered mineral)	Hardness	Lustre
Calcite	Colourless, white	White	3	Dull to vitreous
Quartz	Colourless, purple, red, yellow	White	7	Vitreous

(a) Explain how calcite crystals can be distinguished from quartz crystals.

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(b) Pure quartz crystals are colourless. Explain the large variation in colour indicated in the table.

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Creating a classification key

2. Use the following table to create a written classification key for these minerals.

Mineral	Colour	Streak	Hardness	Lustre
Chalcopyrite	Yellow	Green, black	3.5–4	Metallic
Chlorite	Grey	White, green	2.2–5	Pearly
Galena	Grey	Grey	2.5	Metallic
Haematite	Grey	Red, brown	5–6	Metallic
Magnetite	Black	Black	5.5–6.5	Metallic
Pyrite	Yellow	Green	6–6.5	Metallic

Key:

1A.

1B.

Calcite photo: madllen / 123RF.com
Quartz photo: thomaseder / 123RF.com

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